CartSure

E-Commerce Application

Software Testing Test Plan Document

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Testing Type: Manual & Automation Testing

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1. Introduction

This Test Plan document defines the testing strategy, scope, objectives, schedules, resources, and deliverables for the CartSure E-Commerce Application. The objective is to ensure the application operates efficiently, securely, and provides a smooth shopping experience for end users.

2. Business Scenario

CartSure is a real-time eCommerce platform where users can browse products, search items, add products to cart, proceed to checkout, complete online payments, and receive order confirmations. The QA team is responsible for validating all business-critical functionalities before deployment.

3. Scope of Testing

In Scope:

- Login & Signup Functionality
- Product Listing & Search
- Cart & Checkout Functionality
- Payment Gateway Integration
- Order History & Confirmation
- UI/UX Validation
- Responsive Testing

Out of Scope:

- Third-party payment gateway internal logic
- Logistics backend systems
- Infrastructure & server-side performance tuning

4. Objectives

- Validate complete end-to-end shopping workflow.
- Ensure users can successfully browse and purchase products.
- Verify secure and successful payment processing.
- Ensure application stability and usability.
- Identify and report defects before production release.

5. User Journey Testing

The following user journey will be validated during testing:

1. User opens website
2. User searches for Shoes
3. User views product details
4. User adds product to cart
5. User proceeds to checkout
6. User enters shipping details
7. User completes payment
8. User receives order confirmation

6. Test Strategy

Testing activities will follow the Software Testing Life Cycle (STLC), including requirement analysis, test planning, test case development, environment setup, test execution, defect reporting, regression testing, and final sign-off.

7. Testing Types

Functional Testing: Verify all application functionalities.

UI Testing: Validate layout, design, and responsiveness.

Regression Testing: Ensure existing functionalities work after updates.

Integration Testing: Validate module interactions.

Compatibility Testing: Verify functionality across browsers and devices.

Performance Testing: Validate system performance under load.

Security Testing: Validate authentication and input handling.

Automation Testing: Automate critical workflows using Selenium and TestNG.

8. Functionality Testing Checklist

Home Page: Navigation menu, featured products, banners.
Product Page: Product title, images, filters, sorting.
Cart Module: Add/remove/update cart items.
Checkout: Address form validation and order summary.
Payment: Payment success and failure scenarios.
Information Pages: Privacy Policy, Returns Policy, About Page.

9. Test Environment

Operating Systems: Windows 10/11
Browsers: Chrome, Firefox, Edge
Devices: Desktop, Tablet, Mobile
Automation Tools: Selenium, TestNG
Defect Tracking Tools: Jira/Bugasura

10. Entry & Exit Criteria

Entry Criteria:

• Requirements finalized
• Build deployed successfully
• Test environment ready
• Test data prepared

Exit Criteria:

• All critical test cases executed
• No high severity defects open
• Regression testing completed
• Final report approved

11. Resource Planning

QA Lead – Test planning and monitoring
Tester – Test case execution and defect reporting
Automation Engineer – Automation script development

Developer – Bug fixing and technical support
Business Analyst – Requirement clarification

12. Testing Schedule

Requirement Analysis – 2 Days
Test Planning – 2 Days
Test Case Design – 4 Days
Environment Setup – 2 Days
Test Execution – 7 Days
Regression Testing – 3 Days
Final Test Report – 1 Day

13. Test Deliverables

• Test Plan Document
• Test Strategy Document
• Test Scenarios & Test Cases
• Requirement Traceability Matrix (RTM)
• Defect Reports
• Automation Scripts
• Final Test Summary Report

14. Assumptions & Dependencies

Assumptions: Stable environment, finalized requirements, and available test data.

Dependencies: Application build availability, developer support, and payment gateway sandbox access.

15. Defect Management Process

All identified defects will be logged into Jira/Bugasura with appropriate severity and priority levels. After fixes are provided by developers, QA team will perform retesting and regression testing.

16. Tools Used

• Selenium – Automation Testing
• TestNG – Test Framework
• Jenkins – CI/CD Integration
• Apache JMeter – Performance Testing
• Jira/Bugasura – Defect Tracking
• GitHub – Version Control

17. Approval

This document requires approval from QA Lead, Project Manager, and Client Stakeholders before testing execution begins.